

Iniziato	27 giugno 2025, 11:10
Stato	Completato
Terminato	27 giugno 2025, 12:25
Tempo impiegato	1 ora 14 min.
Valutazione	11,80 su un massimo di 18,00 (66%)
Riepilogo del tentativo	1 2 3 4 5 6 7 8 9 10 11

Domanda 1

Risposta
corretta

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalty for a wrong answer)

The attention for matrices Q, K, V (queries, keys, values) is computed as follows:

$$Attention(Q, K, V) = softmax(\frac{QK^T}{\sqrt{d_k}})V$$

Where d_k is the dimensionality of the keys.

The following are the two input tokens that reach an attention layer:

$$x = \begin{bmatrix} 0 & 0 & 0 \\ 4 & 0 & 0 \end{bmatrix}$$

The following are the W_q , W_k , W_v matrices for the attention layer:

$$W_q = \begin{bmatrix} -2 & -1 \\ 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$W_k = \begin{bmatrix} 0 & -2 \\ -1 & 0 \\ -1 & 1 \end{bmatrix}$$

$$W_v = \begin{bmatrix} 0 & 1 & -2 \\ 1 & 0 & 0 \\ -1 & -1 & -1 \end{bmatrix}$$

Compute the output of the attention operation for the first token.

Write the answer in the following boxes (one dimension for each box).

Report at least 4 significant figures.

Dimension 1: ✓

Dimension 2: ✓

Dimension 3: ✓

1 point (-25% penalty for each wrong answer)

Domanda 2

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

Which of the following best describes how a true positive is identified when using cosine similarity with a threshold

- ☐ a. An actual requirement is a true positive if at least one extracted requirement has a cosine similarity below T
- ☐ b. Every extracted requirement is a true positive as long as it appears in the same paragraph as the actual requirement.
- ☐ c. A true positive is any extracted requirement whose cosine similarity is below T for all actual requirements.
- ☐ d. An extracted requirement is a true positive if it exactly matches the wording of any actual requirement.
- ☒ e. An extracted requirement is a true positive if its cosine similarity with at least one actual requirement exceeds T, and it is the one with the highest similarity among possible matches. ✓

Risposta corretta.

La risposta corretta è:

An extracted requirement is a true positive if its cosine similarity with at least one actual requirement exceeds T, and it is the one with the highest similarity among possible matches.

Domanda 3

Risposta
errata

Punteggio
ottenuto -0,15
su 1,00

1 point (-15% penalty for a wrong answer)

What is the main novelty of In-Context Learning?

- ☐ a. It encodes the task in a prompt using a separate encoder network
- ☐ b. It allows achieving better performance than fine-tuning based approaches
- ☐ c. It improves generalization by updating the model's internal memory
- ☐ d. It allows to perform new tasks without parameter updates
- ☐ e. It trains a smaller model to mimic a larger teacher model
- ☐ f. It stores the examples in a retriever-based memory module
- ☐ g. It fine-tunes the model weights based on the provided examples
- ☒ h. None of the other answers is correct ✗

Risposta errata.

La risposta corretta è:

It allows to perform new tasks without parameter updates

Domanda 4

Risposta
corretta

Punteggio
ottenuto 1,00
su 1,00

1 point (-25% penalty for each wrong answer)

Consider the following prompt:

Write a report about a football match between [choose two teams: e.g., a national team and a club, two rival schools, or two fictional teams]. The match takes place at [choose a stadium or setting: e.g., a packed Olympic stadium, a muddy local field, or a futuristic arena]. The game begins with [describe the opening: e.g., early pressure from one side, a surprise goal, or a controversial decision]. Highlight the performance of key players like [insert a standout player or two] and describe how their actions shaped the game. Include

commentary on [mention key events: e.g., a turning point goal, a red card, a missed penalty]. The match ends with [state the final score and its implications], and the report concludes with [analysis: e.g., coach reactions, what this means for the season, crowd reactions].

What type of prompt is it?

- ☐ a. Tabular prompting
- ☐ b. Few-shot
- ☐ c. Short answer
- ☐ d. Creative writing
- ☒ e. Fill-in-the-blank ✓

Risposta corretta.

La risposta corretta è:
Fill-in-the-blank

Domanda 5

Completo

Punteggio
ottenuto 1,00
su 2,00

2 points (no penalty for wrong answers)

You're evaluating an LLM that extracts functional requirements from a specification for a Workforce Management System (WFM). The LLM is given a paragraph of system requirements and produces a list of extracted candidate requirements.

"Managers can assign tasks to employees and monitor their progress through dashboards. The system allows exporting weekly timesheets and sending automated reminders for pending approvals. Employees must be able to request vacation days via the portal. Integration with external payroll systems is required. The system should support multi-language user interfaces and allow role-based access control."

From this, the ground truth requirements are:

- G1: Assign tasks to employees
- G2: Monitor task progress via dashboards
- G3: Export weekly timesheets
- G4: Send automated approval reminders
- G5: Request vacation days via the portal
- G6: Integrate with external payroll systems
- G7: Support multi-language user interfaces
- G8: Allow role-based access control

Requirements extracted by the LLM:

- L1: Assign work items to staff
- L2: View performance dashboards
- L3: Export timesheet data
- L4: Send automated approval reminders
- L5: Notify managers about task deadlines
- L6: Share dashboards with other departments
- L7: Track overtime hours
- L8: Enable vacation requests from mobile app
- Identify the following:
 - True Positives (TP)
 - False Positives (FP)
 - False Negatives (FN)

Compute:

- Precision = $TP / (TP + FP)$
- Recall = $TP / (TP + FN)$
- F1-Score = $2 \times (P \times R) / (P + R)$

You can write in free text your assumptions about the matching between ground truth requirements and LLM-extracted requirements

L1: FN
L2: FN
L3: FN
L4: TP
L5: FP
L6: FP
L7: FP
L8: FP

Precision = $1/5 = 20\%$

Recall = $1/4 = 25\%$

F1-Score = 20%

Commento:

False negatives must be considered starting from the ground truth, i.e., the requirements in the ground truth that are not extracted by the LLM. Assumptions about the others from L4 to L8 are correct

Domanda 6

Completo

Punteggio
ottenuto 3,00
su 3,00

3 points (no penalty for wrong answers)

You are designing a system of LLAMA-based agents to automatically write human-readable bug descriptions for software issues based on:

- The GitHub issue (title, comments, labels)

Example: Title: Font resets after screensaver; Comments: Happens when using StyledText widget after resume; Labels: SWT, UI

- The git diff associated with the fix (example below)

```
git diff 68f73a31d3bd23bb9be3de8de4cfa69258483b46
+++ b/[...]/org.eclipse.swt/widgets/Composite.java
[empty lines omitted]
- void updateFont (Font oldFont, Font newFont) {
+ boolean updateFont (Font oldFont, Font newFont) {
    Control [] children = _getChildren ();
    for (int i=0; i<children.length; i++) {
        Control control = children [i];

+++ b/[...]/org.eclipse.swt/widgets/Control.java
[empty lines omitted]
- void updateFont (Font oldFont, Font newFont) {
-     Font font = getFont ();
-     if (font.equals (oldFont)) setFont (newFont);
+ boolean updateFont (Font oldFont, Font newFont) {
+     boolean sameFont = getFont ().equals (oldFont);
+     if (!sameFont) setFont (newFont);
+     return !sameFont;
+ }
void updateLayout (boolean resize, boolean all) {
```

- The full source code file in which the git diff occurred

Design an agentic pipeline of LLAMA agents to generate a concise and informative bug description in natural language. The final description should explain:

- The observed behavior
- The expected behavior
- The likely root cause, inferred from the code diff and source

For each agent, write the prompt including System, context, possible inputs from previous prompts, and expected output.

Notice: it is not necessary to write the python calls to the transformers or code for output parsing.

The architecture that we can use here is a sequential chain of LLM Agents. The first agent should extract the observed behavior given the issue of Git, the second should extract the expected behavior given the piece of code from Git and the third one given the output of the first two agents and the Git diff should extract the likely root cause and display the bug description.

```
prompt1 = """
```

```
You are an expert in testing and debugging for software applications and your task is to write a concise and informative, human-readable, bug description for the input issue. The output that i require to you is the observed behavior of the issue given in input, in text format.
```

```
###Example 1
```

```
**Issue:**
```

```
[issue_ex1]
```

```
**Observed behavior:**
```

```
[behavior_issue1]
```

```
###Example 2
```

```
**Issue:**
```

```
[issue_ex2]
```

```
**Observed behavior:**
```

```
[behavior_issue2]
```

```
###Your task
```

```
Given the issue, write in text format the observed behavior:
```

```
{ISSUE}
```

```
**Observed behavior:**
```

```
"""
```

```
prompt2 = """
```

```
You are an expert in testing and debugging software applications, your task is to write in human-readable text format, the expected behavior of a snippet of code given in input.
```

```
###Example 1
```

```
**Code:**
```

```
[code_ex1]
```

```
**Expected behavior:**
```

```
[behavior_exp1]
```

```
###Example 2
```

```
**Code:**
```

```
[code_ex2]
```

```
**Expected behavior:**
```

```
[behavior_exp2]
```

```
###Your task
```

```
Given the code, write in text format the expected behavior:
```

```
{CODE}
```

```
**Expected behavior:**
```

```

"""

prompt3 = """
You are an expert in testing and debugging for software applications and your task is to write a concise and
informative, human-readable, bug description for the input issue. The output that i require to you is the likely
root cause, inferred from the code diff and source in text format, given in input the observed behavior, the
expected behavior and the git diff related.
###Example 1
**Observed behavior:**
[obs_behavior1]
**Expected behavior:**
[behavior_exp1]
**Git diff:**
[git_diff1]

**Likely root cause:**
[lrc1]

###Example 2
**Observed behavior:**
[obs_behavior2]
**Expected behavior:**
[behavior_exp2]
**Git diff:**
[git_diff2]

**Likely root cause:**
[lrc2]

###Your task
Given the observed behavior, the expected behavior and the git diff, write in text format the likely root cause:
{OBSERVED_BEHAVIOR}
{EXPECTED_BEHAVIOR}
{GIT_DIFF}
**likely root cause:**
"""

```

Commento:

Domanda 7

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

You are given the following sequence of characters.

e e e e e b c a b b b b

Initially, each character is a separate token.

Apply **Byte-Pair Encoding** to produce 3 additional tokens. What are the generated tokens?

Write each token, in the correct order, in the following slots.

Separate the merged tokens with a comma (and no spaces).

Examples:

1) if the merged tokens are aa and b, write: **aa,b**

2) if the merged tokens are x and y, write: **x,y**

1st merge: ✓

2nd merge: ✓

3rd merge: ✓

Domanda 8

Risposta
corretta

Punteggio
ottenuto 1,50
su 1,50

1.5 points (no penalty for a wrong answer)

Apply absmax quantization to the following values x_1 , x_2 , x_3 , x_4 , to map them to int8.

- $x_1 = -7$
- $x_2 = 3$
- $x_3 = -2$
- $x_4 = 0$

Write the quantized versions in the following boxes.

quantized(x_1) = ✓

quantized(x_2) = ✓

quantized(x_3) = ✓

quantized(x_4) = ✓

Domanda 9

Risposta
errata

Punteggio
ottenuto -0,15
su 1,00

1 point (-15% penalty for a wrong answer)

You are given an encoder-only model. What is the total number of parameters (token + positional + transformer layers)?

Note: assume that none of the layers uses a bias term. You can ignore layer normalization parameters.

- Vocabulary size $V=128,000$
- Embedding/model dimension $d=768$
- Positional embeddings: absolute, learned, for 2048 positions
- Number of layers $L=36$
- Number of attention heads $H=12$
- FF-NN projection dimension: 3072
- Head dimension $d_k = d_v = 64$

Choose the correct answer. The results are reported to the closest million.

- ☐ a. 185 M
- ☐ b. 255 M
- ☒ c. 100 M ✗
- ☐ d. 52 M
- ☐ e. None of the other answers is correct
- ☐ f. 110 M
- ☐ g. 355 M
- ☐ h. 302 M
- ☐ i. 333 M

Risposta errata.

La risposta corretta è:
355 M

**Domanda
10**

Completo

Punteggio
ottenuto 2,00
su 2,00

2 points (no penalties for wrong answers)

Consider the excerpts of recommendation letters generated with a Large Language Model reported below:

Gender	Generated Text
Female	She is great to work with, communicates well with collaborators and fans , and always brings an exceptional level of enthusiasm and passion to her performances.
Male	His commitment, skill, and unique voice make him a standout in the industry , and I am truly excited to see where his career will take him next.
Female	She takes pride in her work and is able to collaborate well with others .
Male	He is a true original, unafraid to speak his mind and challenge the status quo .
Female	Her kindness and willingness to help others have made a positive impact on many.
Male	I have no doubt that his experience in the food industry will enable him to thrive in any culinary setting.

Provide a **brief** critical comment about:

- the possible reasons behind the differences in the output;
- the possible implications of such behaviour of generative AI.

1) The main differences that we can observe, are that the recommendation letters generated for the female are focused more on the soft skills and on the behaviors in social context, it looks more like a description that you provide to a friend. In the other hand, for the male, the letters are focused more on highlight that the man is a leader and an absolute expert on field, so it looks more like a real work application description.

2) The possible implications of this behaviour are:

- Information from the web: on LinkedIn for example there are a lot of job description where they require a leadership male figure, so this could influence the generative AI. For the same reason there are a lot of job description where they require a "social" female.
- Social influence: in the working world there are more male than female in a leadership position, so also this could influence a bit the AI.

Commento:

ok. missing some discussion about the implications in the sense of "consequences" of the behaviour

**Domanda
11**

Risposta
errata

Punteggio
ottenuto -0,90
su 2,00

2 points (~15% penalty for each wrong answer)

Which of the following tasks were used for the pre-training of BERT? Select all correct answers.

- ☒ a. Machine Translation ✖
- ☐ b. Image-Text Matching
- ☒ c. Denoising Autoencoding ✖
- ☐ d. Permuted Language Modeling
- ☐ e. Text Summarization
- ☒ f. Causal Language Modeling ✖
- ☐ g. Next Sentence Prediction
- ☐ h. Masked Language Modeling
- ☐ i. Sentence Ordering Prediction

Risposta errata.

Le risposte corrette sono:

Masked Language Modeling,

Next Sentence Prediction